

NORTH SEA COASTAL HERITAGE MAPPING PROJECT

Submerged Cultural Heritage Survey - Field Methodology Guidelines

Hamburg Maritime Museum and Nordmeer Maritime University

HMM-CHM-22 | Version 3.1 | Issued: 15.10.2025

1. Purpose and Applicability

These guidelines govern all survey activities conducted under the North Sea Coastal Heritage Mapping Project, a joint programme of Hamburg Maritime Museum and Nordmeer Maritime University, funded under agreement HMM-CHM-22. The guidelines apply to all personnel involved in fieldwork, data processing, and reporting, whether employed by the museum, the university, or engaged as contractors.

2. Survey Standards

2.1 Spatial coverage: Survey lines shall achieve 100 percent seabed coverage within the designated survey polygon. Line spacing shall be determined by the MBES beam geometry at the planned survey depth. No gaps exceeding 5 percent of a beam width shall be accepted in the final coverage mosaic.

2.2 Positioning: All survey positions shall be recorded in WGS84 geographic coordinates with an accuracy of better than 1.0 metre at the 95th percentile, verified by post-processed differential GPS with reference to a land-based control point.

2.3 Water column measurement: A sound velocity profile shall be measured at the start of each survey line and whenever depth, temperature, or salinity conditions change significantly. The profile data shall be archived and provided with all processed survey deliverables.

3. Feature Recording

All candidate archaeological features identified in the survey data shall be recorded in the project feature register using the classification scheme set out in Annex A. Feature positions shall be reported in WGS84 with an accuracy consistent with the positional standards set out in Section 2.2. Feature records shall include: anomaly class, depth, estimated dimensions, acoustic signature description, and the survey line reference from which the detection was made.

4. Data Management

Data type	Format	Retention	Repository
Raw MBES data	Kongsberg .all or .kmall	Permanent	University research storage
Processed bathymetry	GeoTIFF, 0.25m resolution	Permanent	University research storage
Feature register	GeoPackage + PDF	Permanent	Shared with museum (Box)
Survey reports	PDF/A	Permanent	Project archive + museum
Processed imagery	GeoTIFF + JPEG	Permanent	Shared with museum (Box)
Field logbooks	Scanned PDF	Permanent	University archive

5. Reporting Requirements

A preliminary feature list shall be delivered within 4 weeks of each field season conclusion. A final survey report shall be delivered within 3 months of each field season conclusion. The final report shall include: a survey methodology summary, a feature catalogue, a coverage assessment, and recommendations for follow-up investigation. Copies of all reports shall be filed with Hamburg Maritime Museum and with the relevant heritage authority.

6. Community and Heritage Consultation

All survey activities in waters of recognised Frisian heritage significance shall be notified to the community liaison representative in advance. Where survey results identify features within areas of Frisian cultural interest, a community briefing shall be held before any investigation plan is finalised. The community consultation protocol developed with the Frisian cultural liaison is attached as Annex B and forms part of these guidelines.

7. Health and Safety

All offshore survey operations shall comply with the applicable Maritime Labour Convention provisions and German Berufsgenossenschaft Verkehr (BGV) regulations for maritime workers. A risk assessment specific to each survey season shall be completed and signed by the scientific lead and the vessel master before departure. Dive operations, if any, are governed by the separate Dive Operations Safety Protocol (document NMU-DIVE-4.2).

8. Intellectual Property and Publication

All data and results produced under this project are jointly owned by Hamburg Maritime Museum and Nordmeer Maritime University in accordance with the project agreement HMM-CHM-22. Publication of project results requires the consent of both parties. Publication protocols are set out in the project agreement. The Frisian community consultation protocol includes specific provisions for community review of results before external publication where those results directly concern Frisian heritage sites or oral traditions.

Annex A - Feature Classification Scheme

Code	Class	Definition
H1	High confidence anthropogenic	Hard linear or complex feature, relief over 0.5 m, no natural explanation
H2	Moderate confidence anthropogenic	Hard feature, relief 0.2-0.5 m, anthropogenic origin likely
M1	Mixed: uncertain origin	Feature with characteristics of both natural and anthropogenic origin
N1	Natural feature	Geomorphological or biological feature confirmed or likely
MH	Modern human origin	Feature consistent with post-1900 human activity
U	Unclassified	Insufficient data for classification; review required